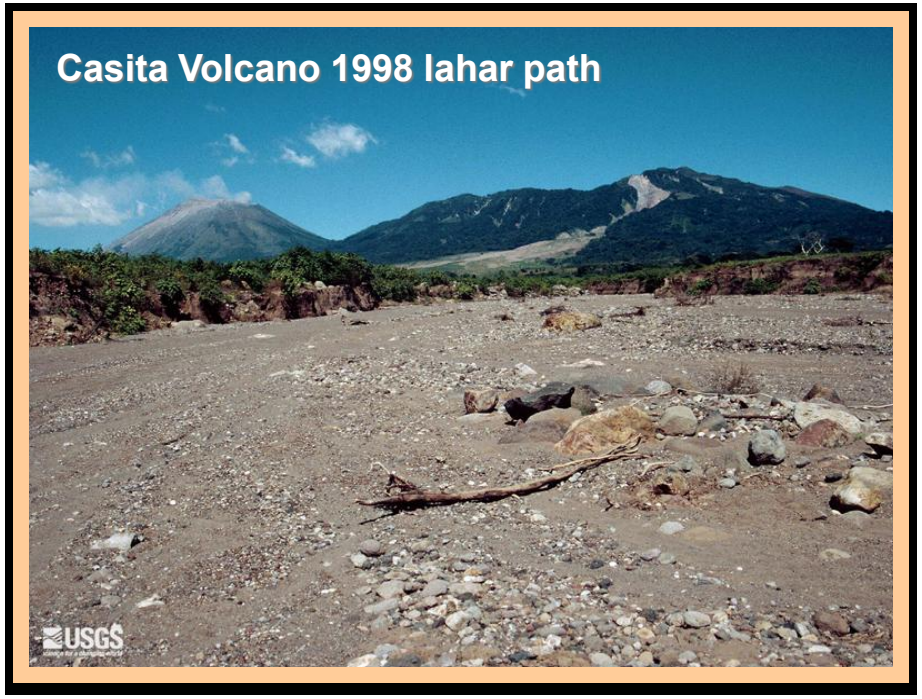
Modeling Lahars



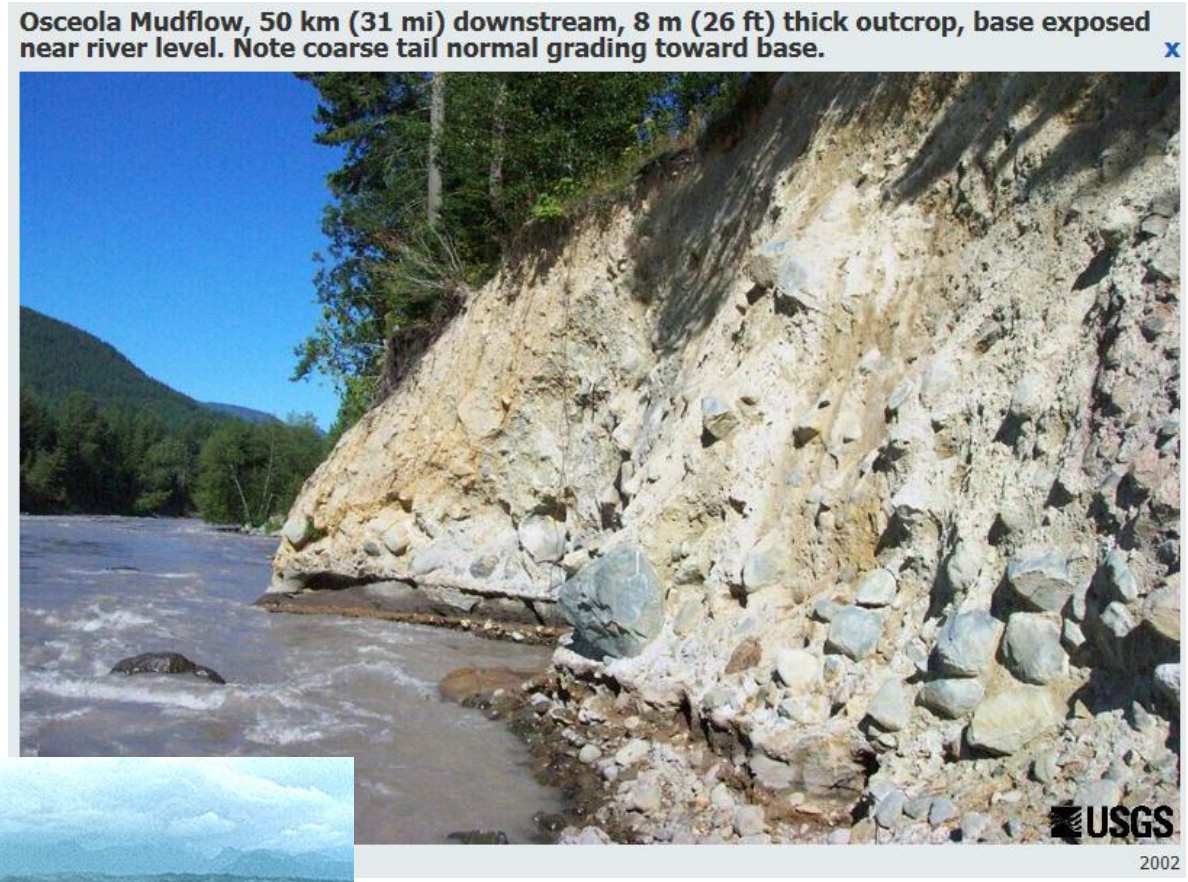


Lahar/debris flow models

- LAHARZ (Iverson et al. 1998): empirical model based on flow volume, inundated area, and maximum cross-sectional area
- Flo2D (OBrien 1993): hydrodynamic model (commercial)
- VolcFlow (Kelfoun & Druitt 2005): single and two-phase flow models
- D-CLAW (George & Iverson 2014): two phase debris flow model with pore-fluid pressure, evolving volume fractions, etc.
- LaharFlow/MultiHaz (Woodhouse et al.) https://www.laharflow.bristol.ac.uk/help/quickstart/
- Grfin Tools (Reid et al. 2025): mostly debris flows

Studying lahars → modeling lahars

1. Field observation and mapping older deposits at Mount Rainier, Washington, USA





2. Experience of lahars and their consequences

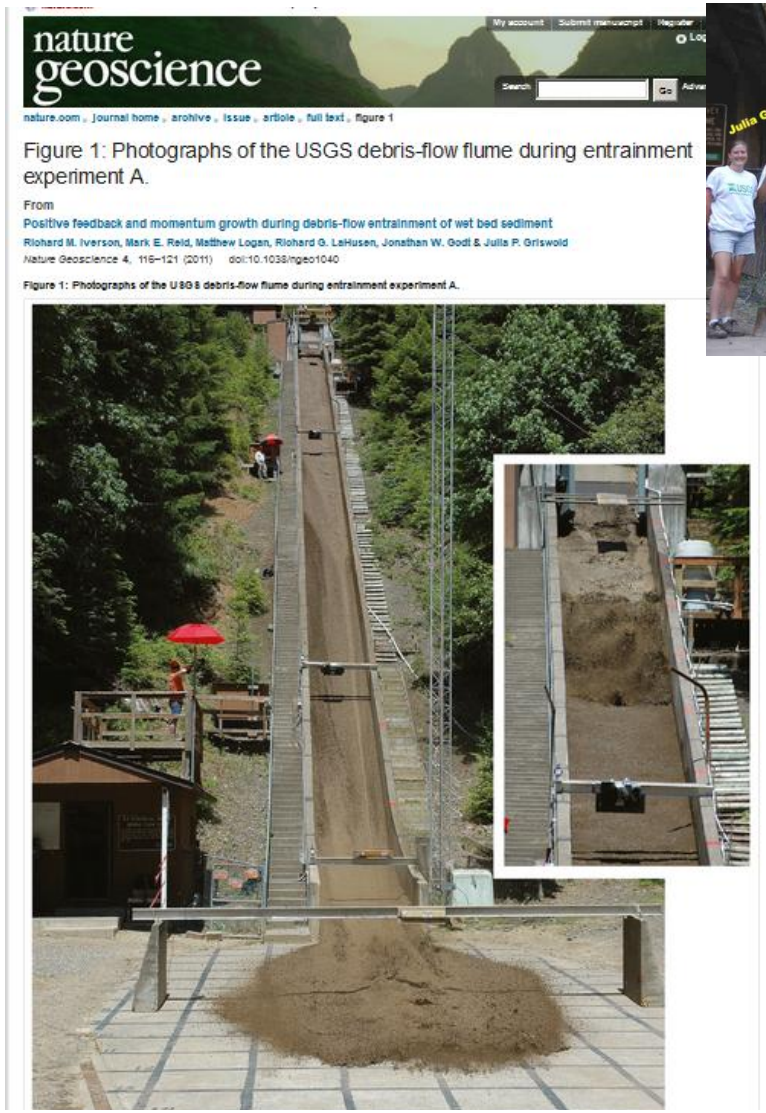
- Nevado del Ruiz; location of the former town of Armero, Colombia
- Pinatubo, Philippines

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3. Experiments to demonstrate field observations; research; confirm modeling; prediction.

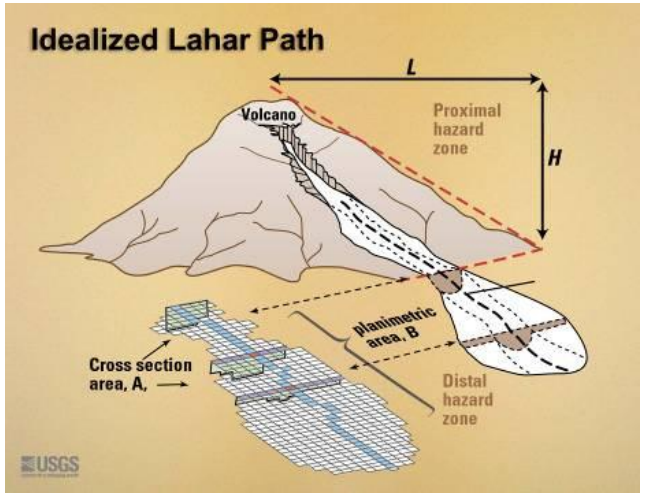
>163 debris flow experiments 1992-2017
>18 hours edited video online
http://pubs.usgs.gov/of/2007/1315/







5. Empirical and Statistical modeling to understand the limits of potential, average inundation and to help make general predictions (LAHARZ)



Dick Iverson, Steve Schilling, Jim Vallance, Julie Griswold

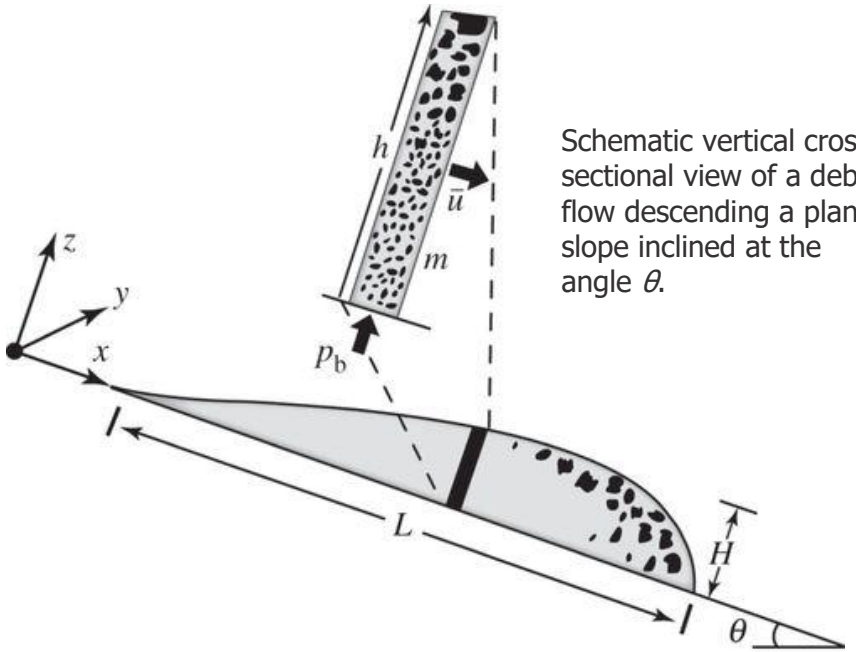


Iverson & Schilling 1998: https://doi.org/10.1130/0016-7606(1998)110%3C0972:ODOLIH%3E2.3.CO;2
Schilling 1998: https://doi.org/10.3133/ofr98638
Griswold & Iverson 2007: https://pubs.usgs.gov/sir/2007/5276/
Schilling 2014: https://pubs.usgs.gov/of/2014/1073/

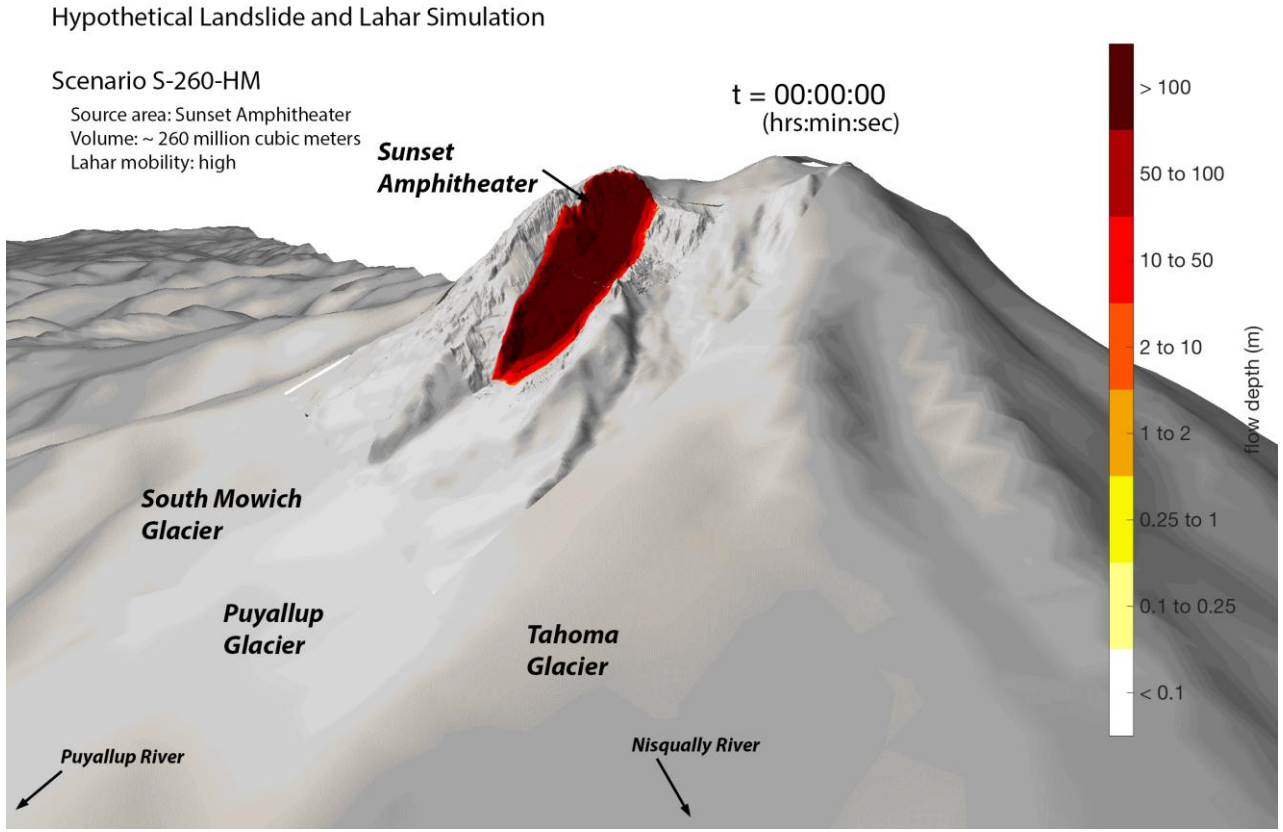


LaharZ model (left) of the possible lahar hazards from the Carbon and Puyallup River drainages from Mount Rainier. The town of Orting, Washington (photograph) sits in the lahar hazard zone.
https://www.usgs.gov/media/images/laharz-model-possible-lahar-hazards-carbon-and-puya

4. Mathematical modeling to understand the dynamics and to make specific predictions (D-CLAW)



Schematic vertical cross-sectional view of a debris flow descending a planar slope inclined at the angle θ.



George et al. 2021 https://pubs.usgs.gov/publication/ofr20211118

David George, Mary Benage, Richard Iverson, Roger Denlinger



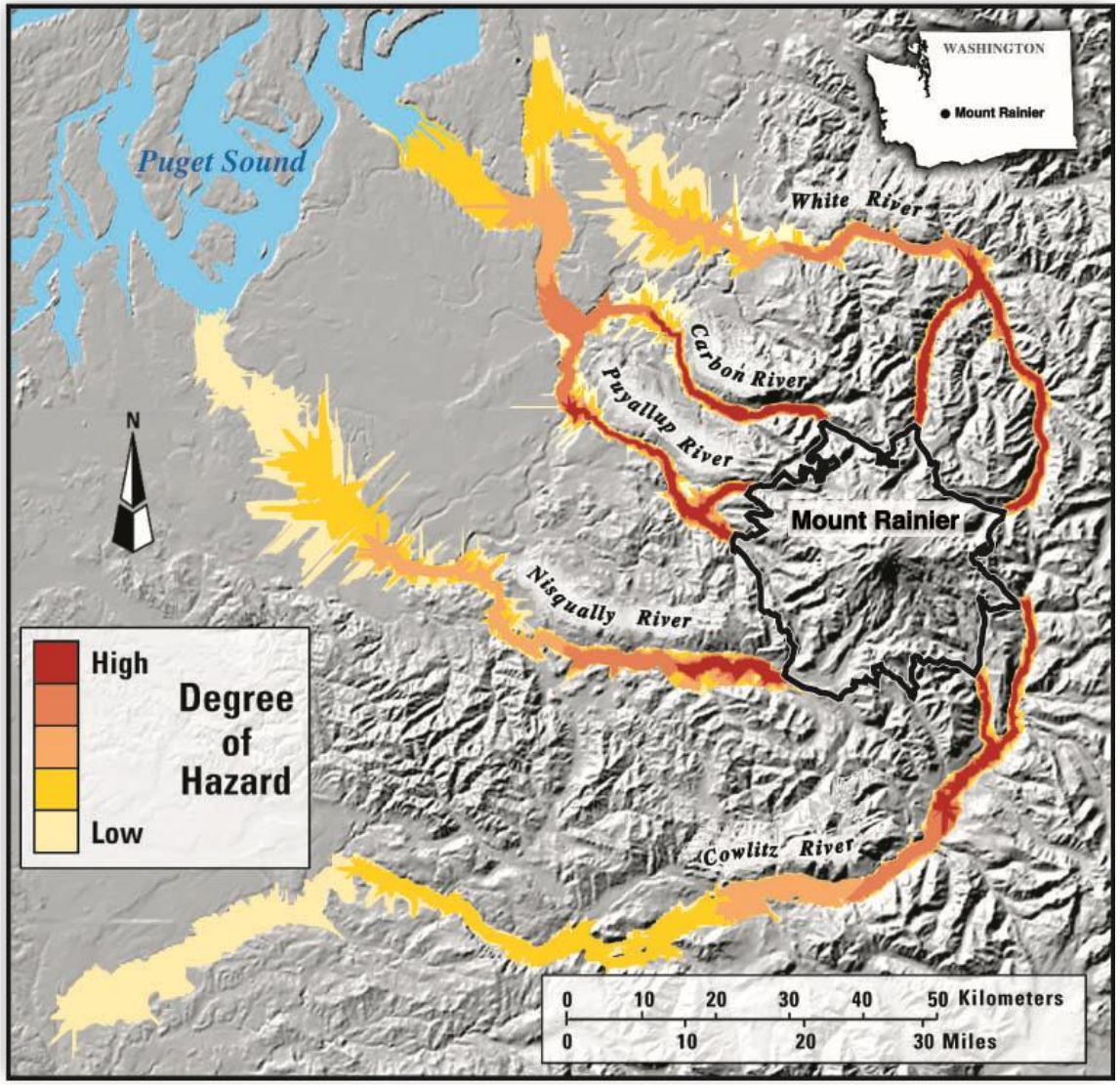
George & Iverson 2014: https://doi.org/10.1098/rspa.2013.0820
Iverson & George 2014: https://doi.org/10.1098/rspa.2013.0819

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About LAHARZ

- Semi-empirical model for forecasting lahar inundation
- First developed by the USGS in 1998 (Iverson et al. 1998, Schilling 1998) in ArcMap
- Updated to ArcMap python tools for new versions of ArcGIS (Schilling 2014)
- Updated for ArcPro in 2023 by Julie Griswold
- Taught at CSAV by Steve Schilling and Julie Griswold since ????
- I started teaching the course with Julie in 2016 as Steve was retiring
- LAHARZ is open-source, but ArcGIS is not
- We've always wanted an open-source version of LAHARZ because ArcGIS is $$$$$$$$$



About LAHARZ

- Keith Blair at University of Bristol wrote a stand-alone Python version in late 2022, with DEM preparation done in QGIS
- I've been working with Keith to suggest new features and improvements over the last 2 years
- You can download this version and install on your local machine from here: https://github.com/Keith1815/laharz
- Keith's Python LAHARZ was installed on the VICTOR platform in 2024, which makes it easy to run without having to worry about installing packages



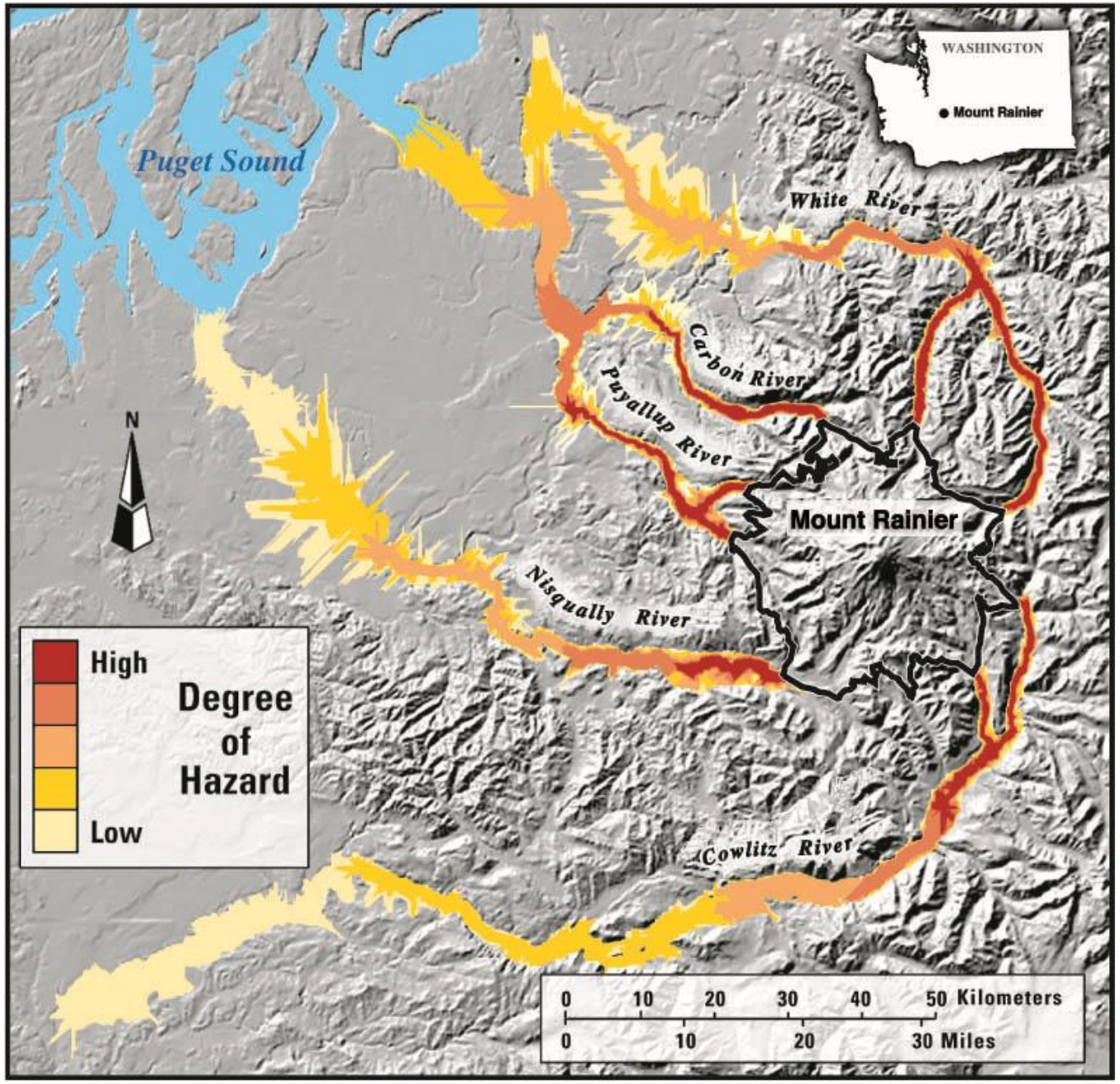
The 2i2c JupyterHub for VICTOR



LAHARZ

Goal: a rapid, objective, reproducible method for forecasting lateral and distal inundation limits that define hazard zones

Guiding Philosophy: Global data on areas inundated by past lahars provide a basis for forecasting future inundation limits at particular locations



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LAHARZ principles

1) inundation by past lahars provides a basis for predicting inundation by future lahars

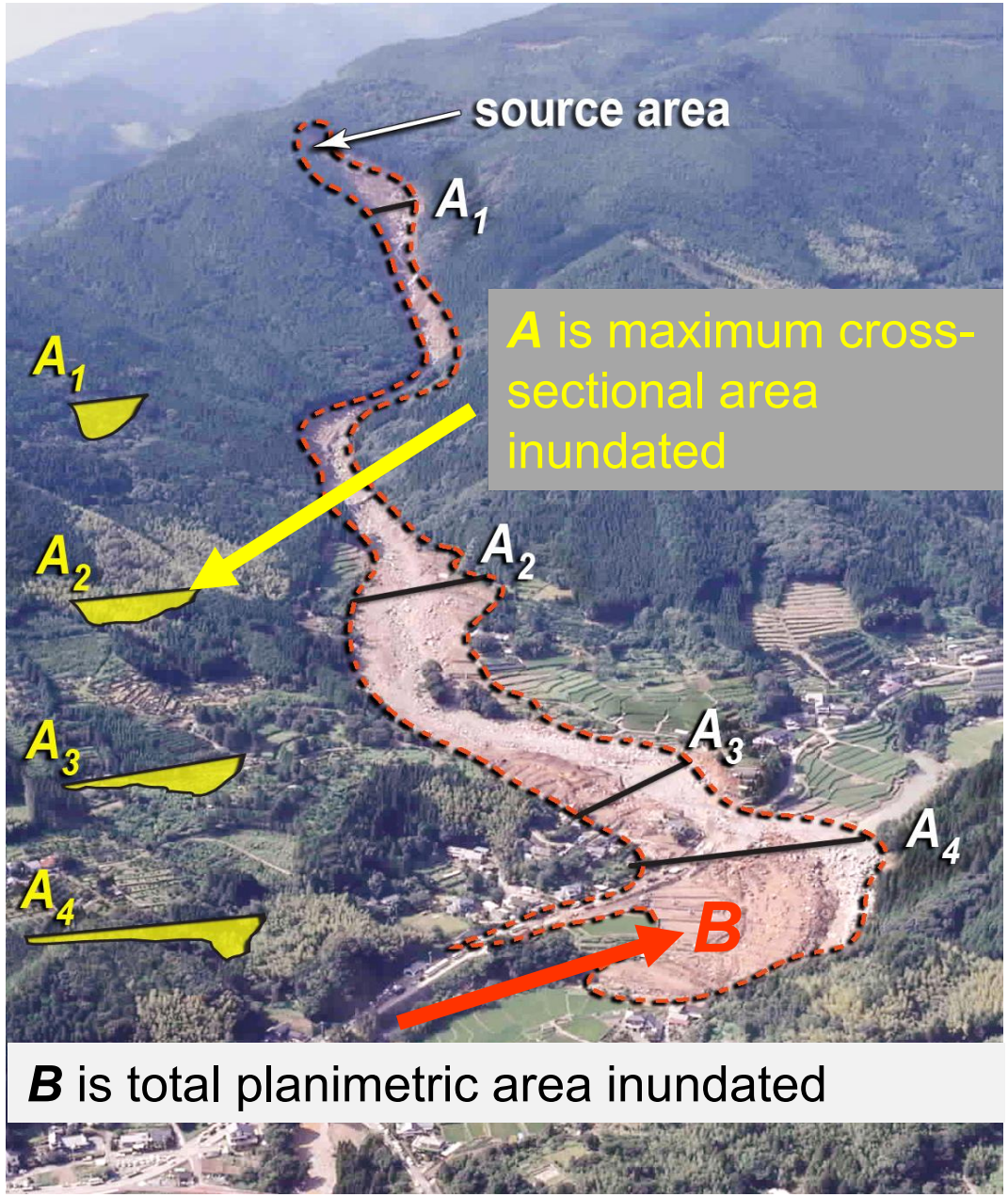
2) distal lahar hazards are confined to valleys that start on volcano flanks

3) lahar volume largely controls the extent of inundation downstream

4) voluminous lahars occur less often than small lahars

5) no one can foretell the exact size of the next lahar to descend a given drainage

(Iverson et al., 1998).

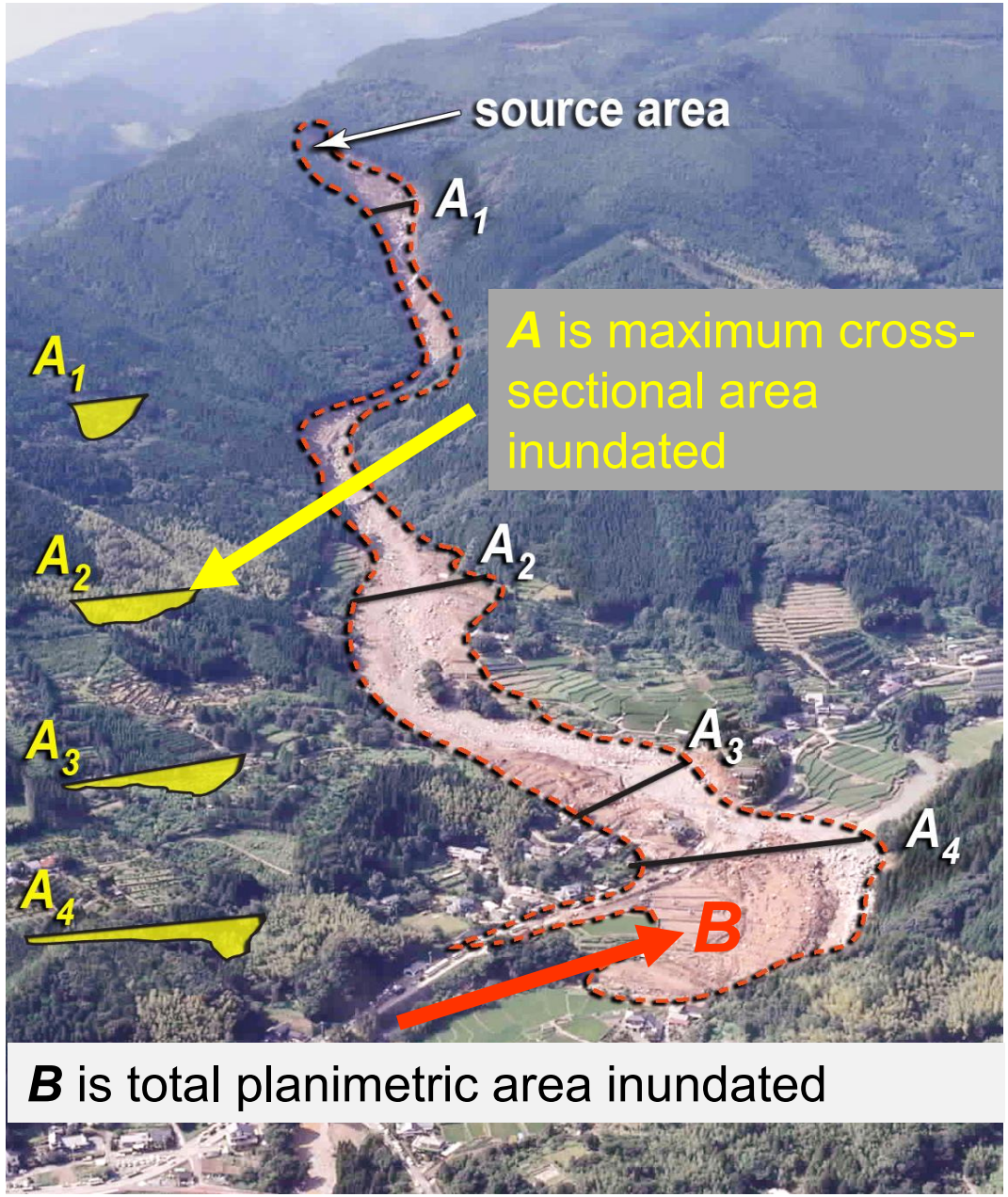


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LAHARZ equations

- From analysis of lahar kinematics, cross-sectional (A) and planimetric (B) area of inundation are proportional to flow volume $(V)^{2/3}$
- ***Max cross-sectional area (A)*** = $c_1 V^{2/3}$
- ***Total planimetric area (B)*** = $c_2 V^{2/3}$
- The coefficients c_1 and c_2 are determined by measuring the maximum cross-sectional area, planimetric area, and volume of many flows and performing regression analysis

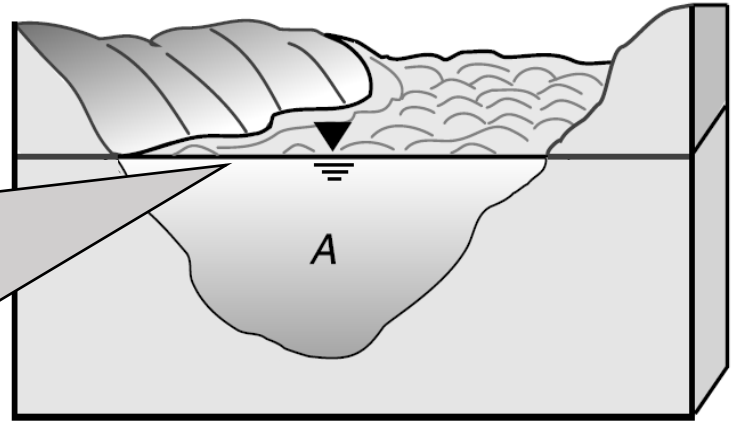


LAHARZ equations: How to estimate coefficients?

- Data are ideally not deposit area, but instead are inundation area and maximum cross-sectional area as determined from high-flow marks, hydrographs, flow observations, etc.

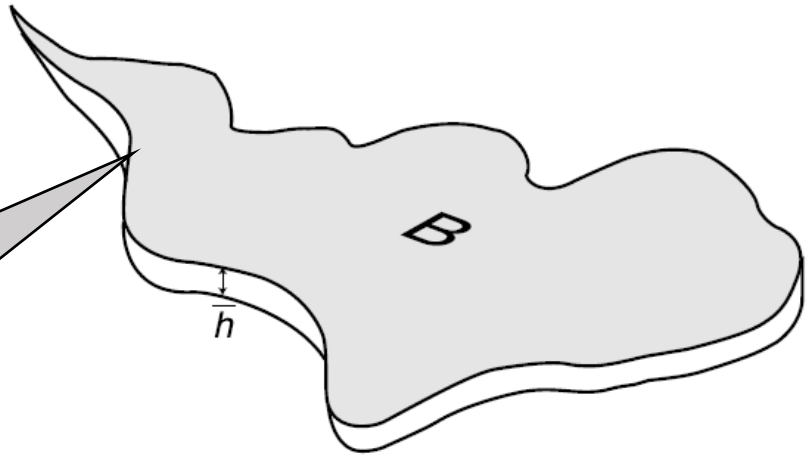
- Data include volumes from analyses of lahar hydrographs or measurements of deposited sediment volumes

high-flow marks indicated by strandlines, levees, embedded gravels, or the height of stripped bark on the trunks of adjacent trees, and the height of log jams.



$$\textit{Max cross-sectional area } (A) = c_1 V^{2/3}$$

lateral limits of any evidence of high-flow lines.



$$\textit{Total planimetric area} = c_2 V^{2/3}$$

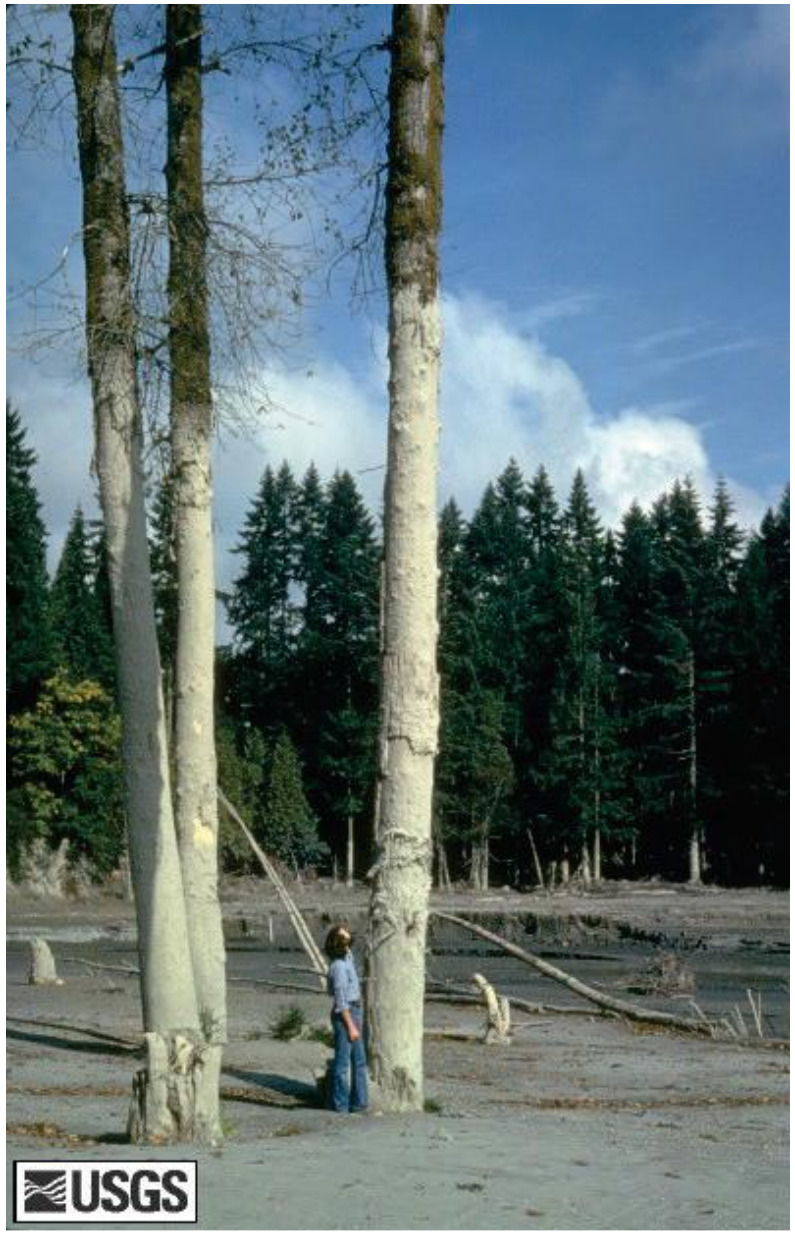
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May 18, 1980 eruption of Mount St. Helens

Photograph taken by Lyn Topinka, U.S. Geological Survey, 1980

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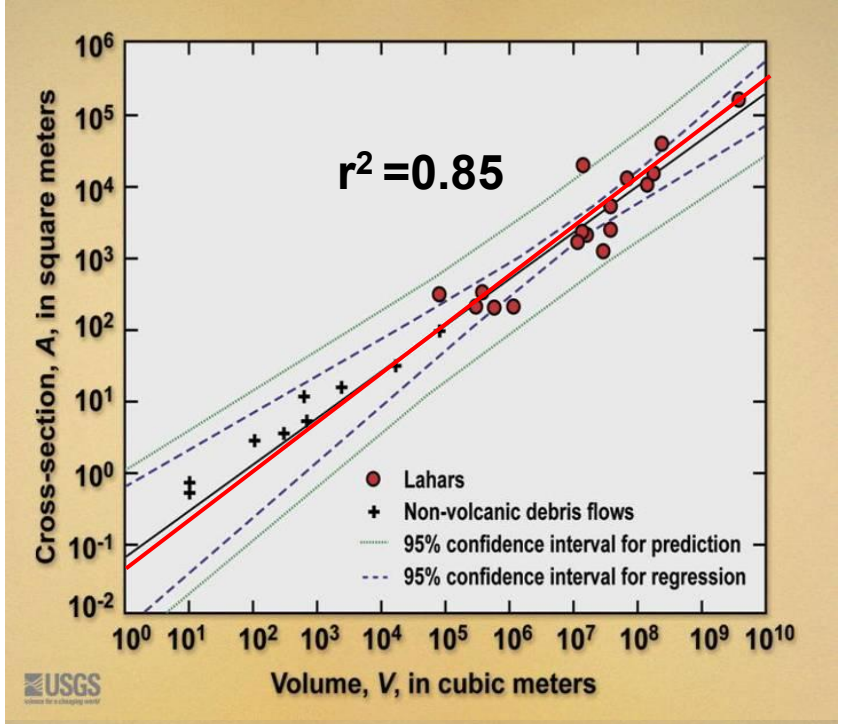
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Takahashi, 1981

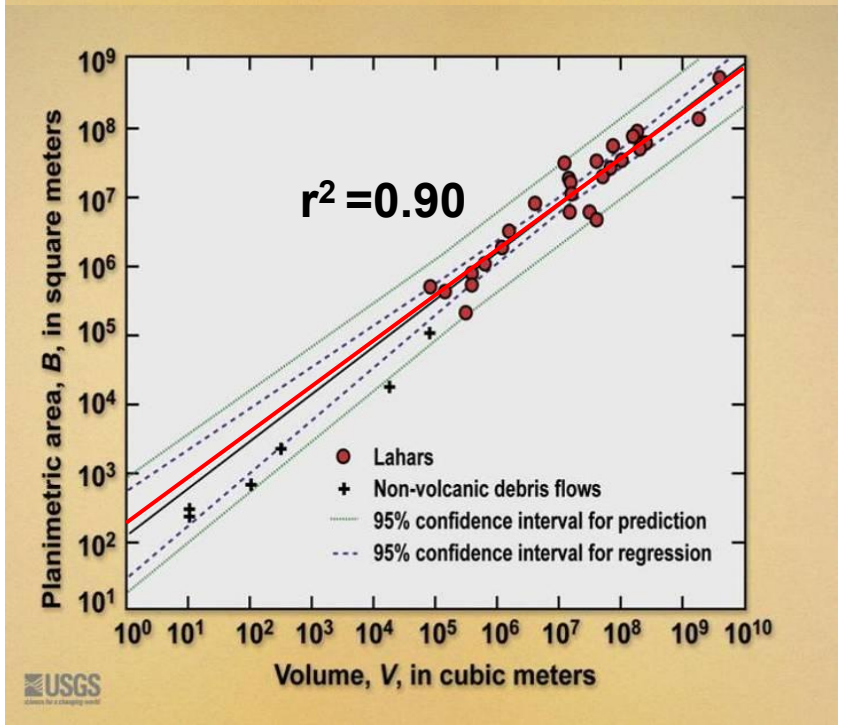


LAHARZ equations: How to estimate coefficients?

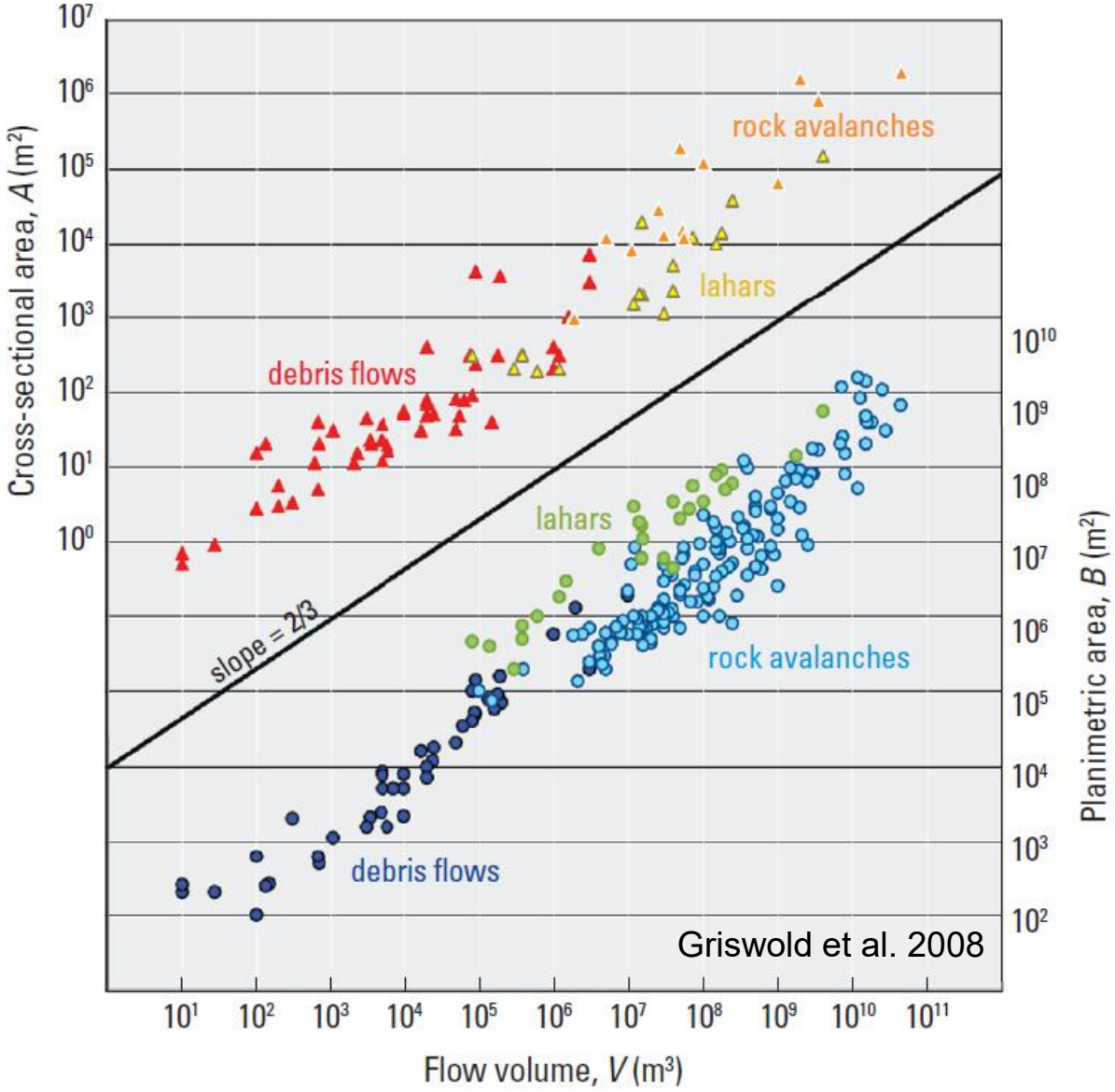
- Cross-sectional (A) and planimetric (B) area of inundation are proportional to flow volume $(V)^{2/3}$
- The coefficients c_1 and c_2 are determined by measuring the maximum cross-sectional area, planimetric area, and volume of many flows and performing regression analysis
- Different flow types (lahars, rockfalls, debris flows) have different coefficients



Cross-sectional area = $c_1 V^{2/3}$

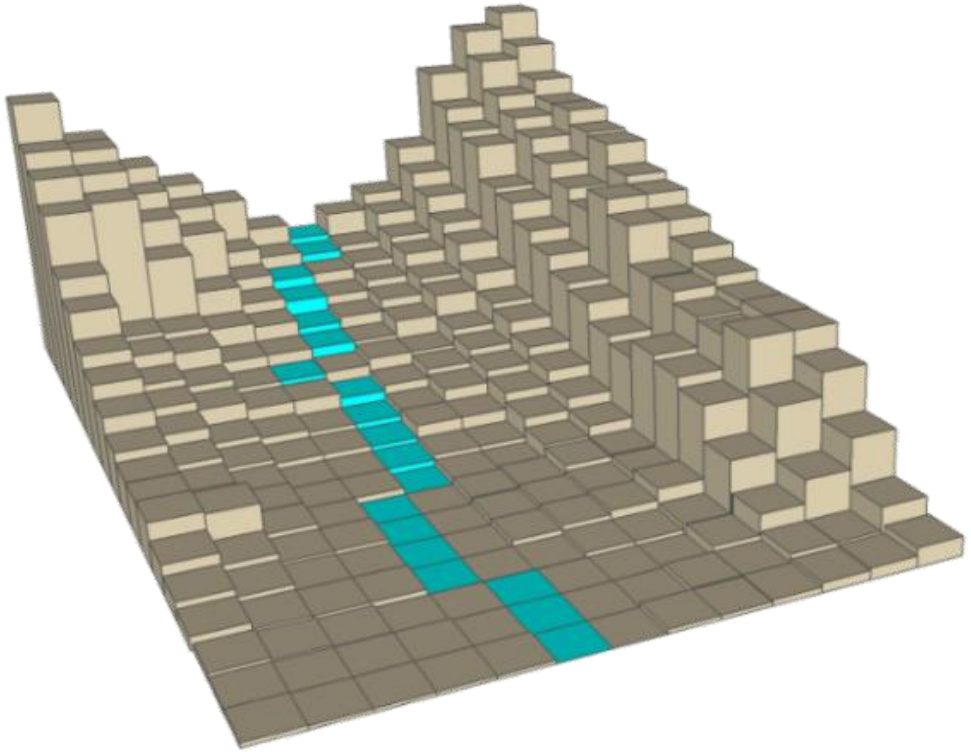


Planimetric area = $c_2 V^{2/3}$



Flow Type	C_1	C_2
Lahars	0.05	200
Debris flows	0.1	20
Rockfalls	0.2	20

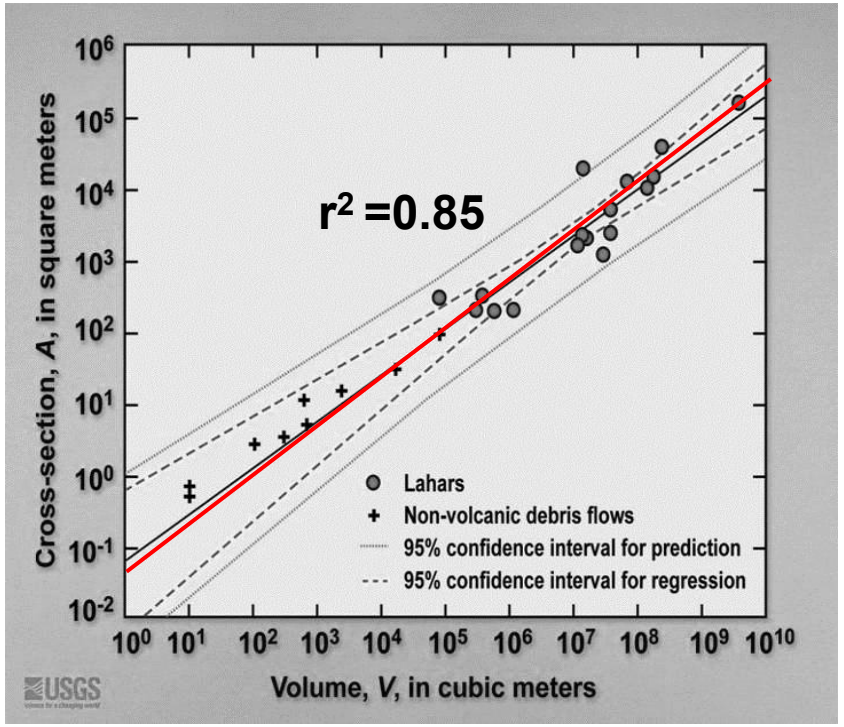
How does LAHARZ work?



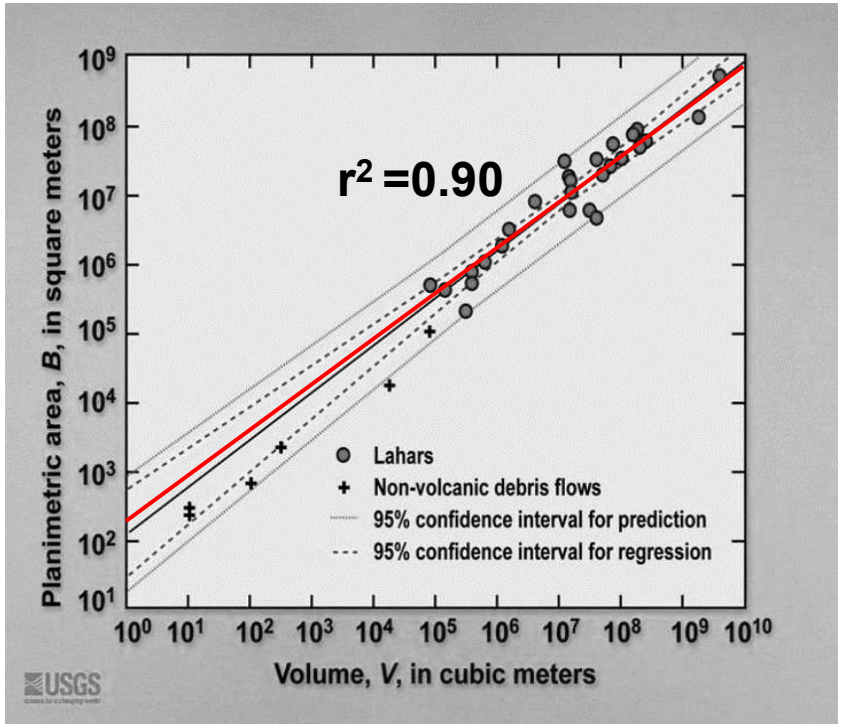
Use LAHARZ to fill sinks in topography, create a hillshade, and produce a stream network of valley thalweg cells.

Specify the volume, V, used for the simulation.

How does LAHARZ work?



***Cross-sectional area* = 0.05 $V^{2/3}$**



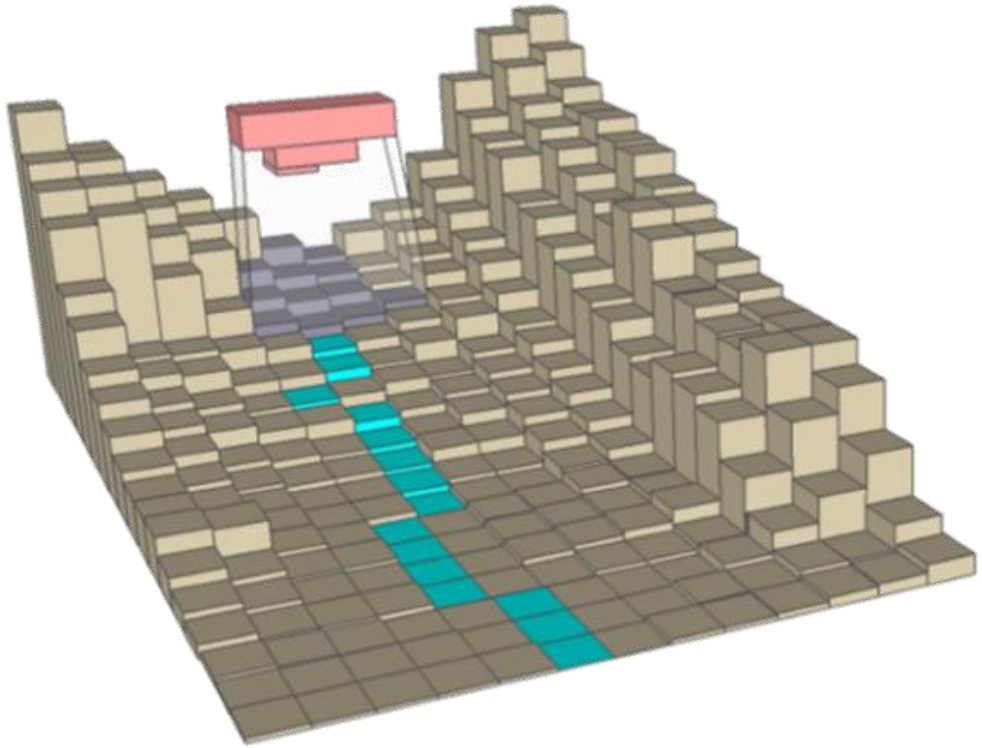
***Planimetric area* = 200 $V^{2/3}$**

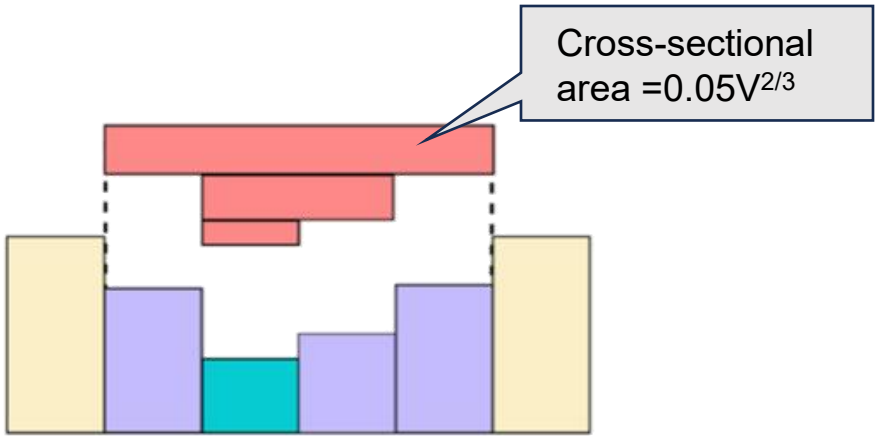
From collected deposit data, determine the coefficients c_1 and c_2 for
Cross-sectional area = $c_1 V^{2/3}$
Planimetric Area = $c_2 V^{2/3}$

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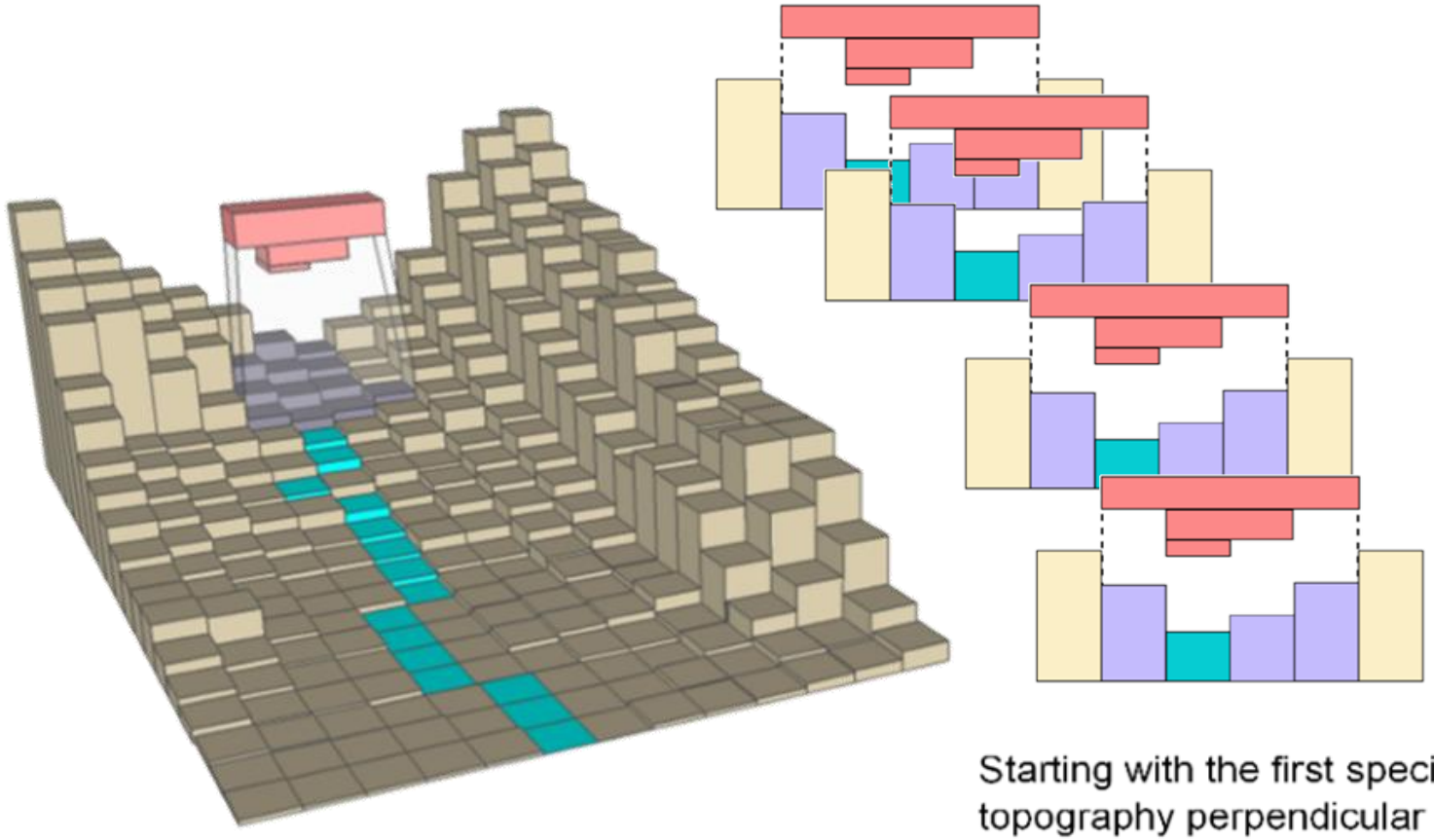
How does LAHARZ work?





Starting with the first specified valley thalweg cell, fill topography perpendicular to the thalweg until cross-sectional area is equal to $C_1V^{2/3}$. Add inundated cells to running planimetric area total, and move to next valley thalweg cell.

How does LAHARZ work?

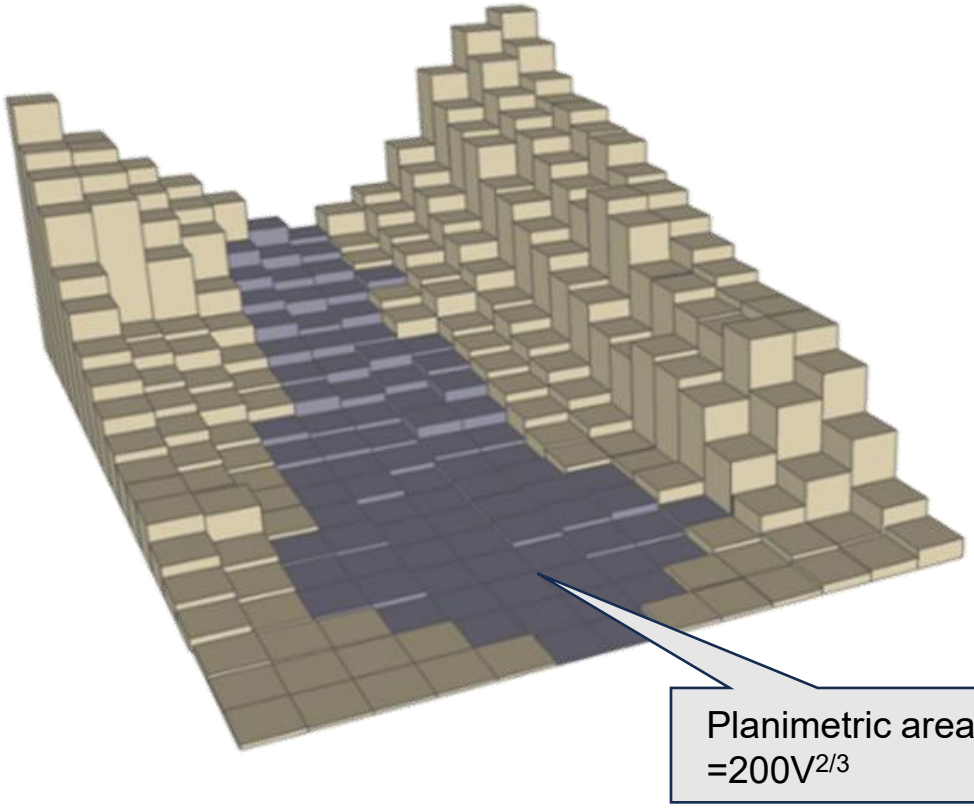


Starting with the first specified valley thalweg cell, fill topography perpendicular to the thalweg until cross-sectional area is equal to $C_1V^{2/3}$ Add inundated cells to running planimetric area total, and move to next valley thalweg cell.

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How does LAHARZ work?



When the running planimetric area total is equal to $C_2V^{2/3}$ stop the simulation.

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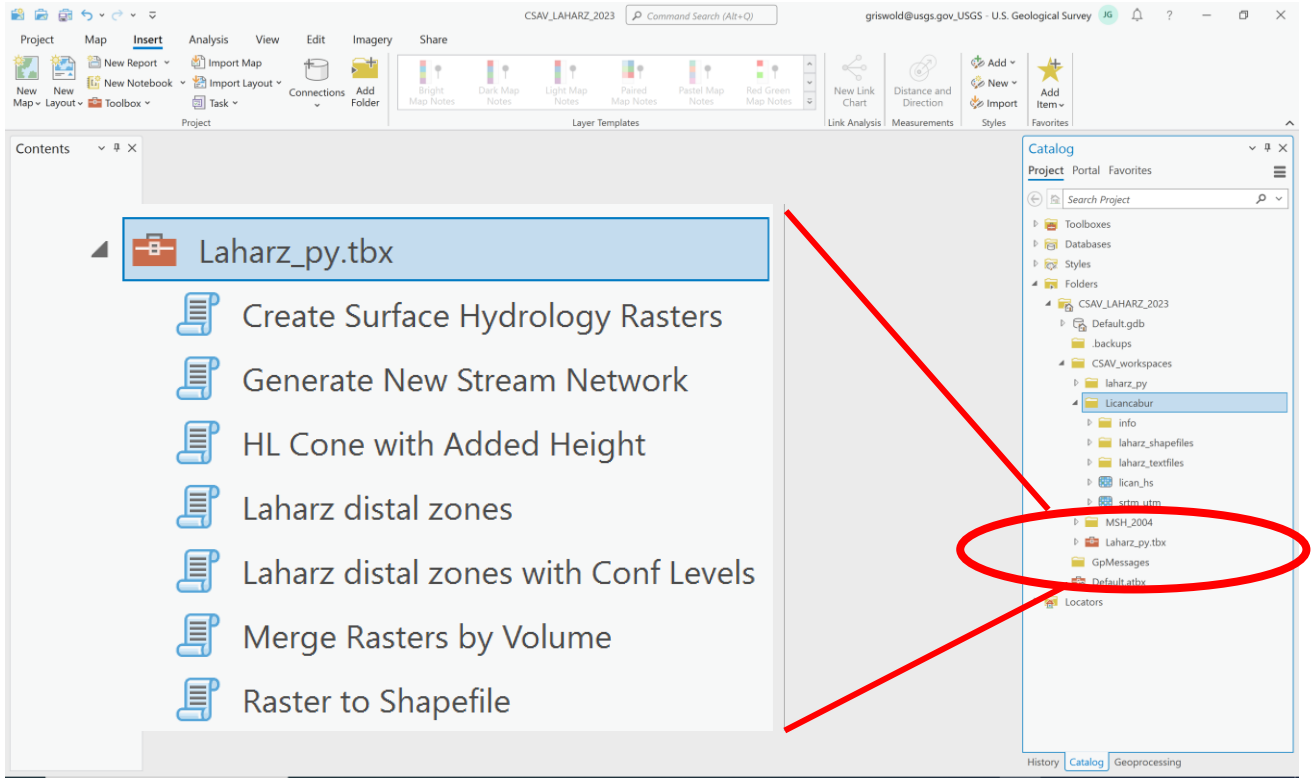
LAHARZ Versions

- ArcGIS ArcMap 10.x
- Laharz_py ArcGIS toolbox
- Schilling et al. 2014
- https://pubs.usgs.gov/of/2014/1073/
- CSAV file folders



LAHARZ Versions

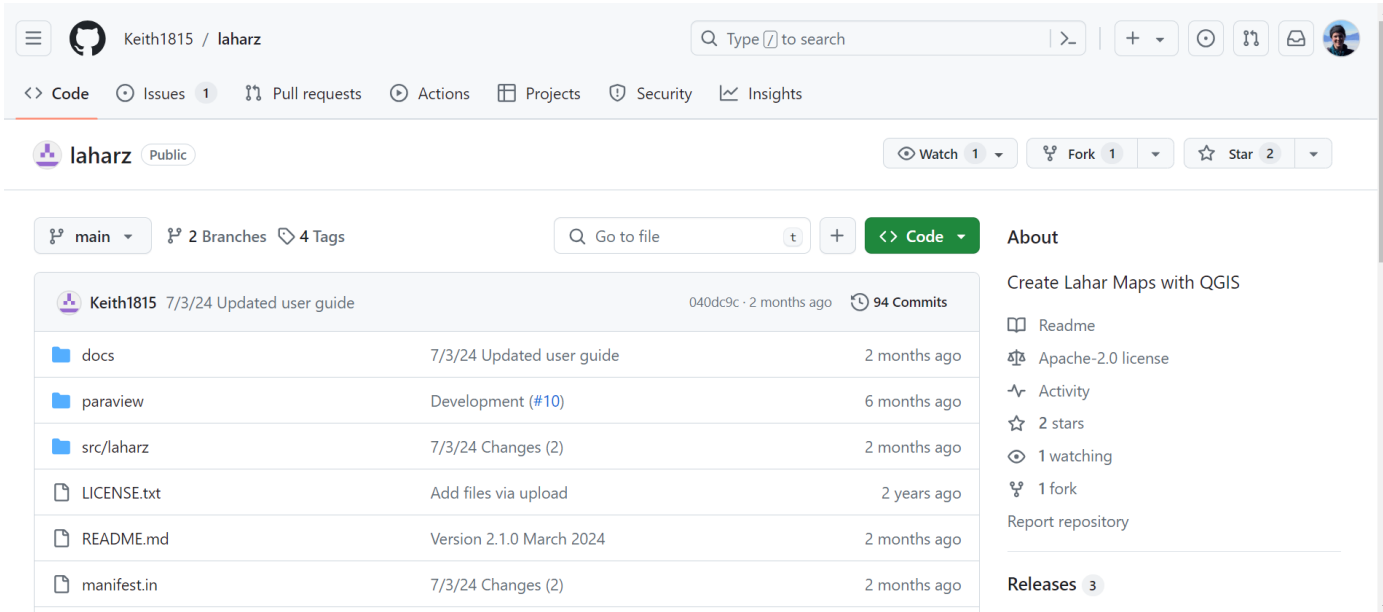
- ArcGIS Pro
- Julie Griswold 2023 rewrite of Schilling et al. 2014 code for ArcGIS Pro
- Email me!
- CSAV file folders



LAHARZ Versions

- Keith Blair stand-alone python version
- https://github.com/Keith1815/laharz





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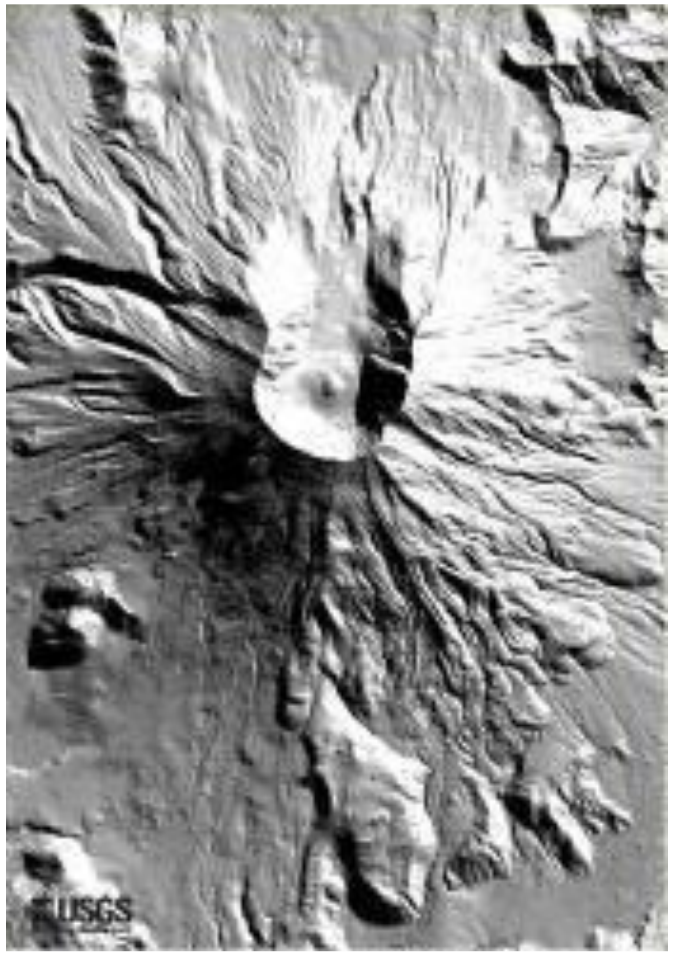
LAHARZ Versions

- VICTOR platform (using QGIS and python-version of LAHARZ by Keith Blair)
- https://victor.2i2c.cloud/
- Nice because you don't have to worry about installing python dependencies, so we'll use it for class



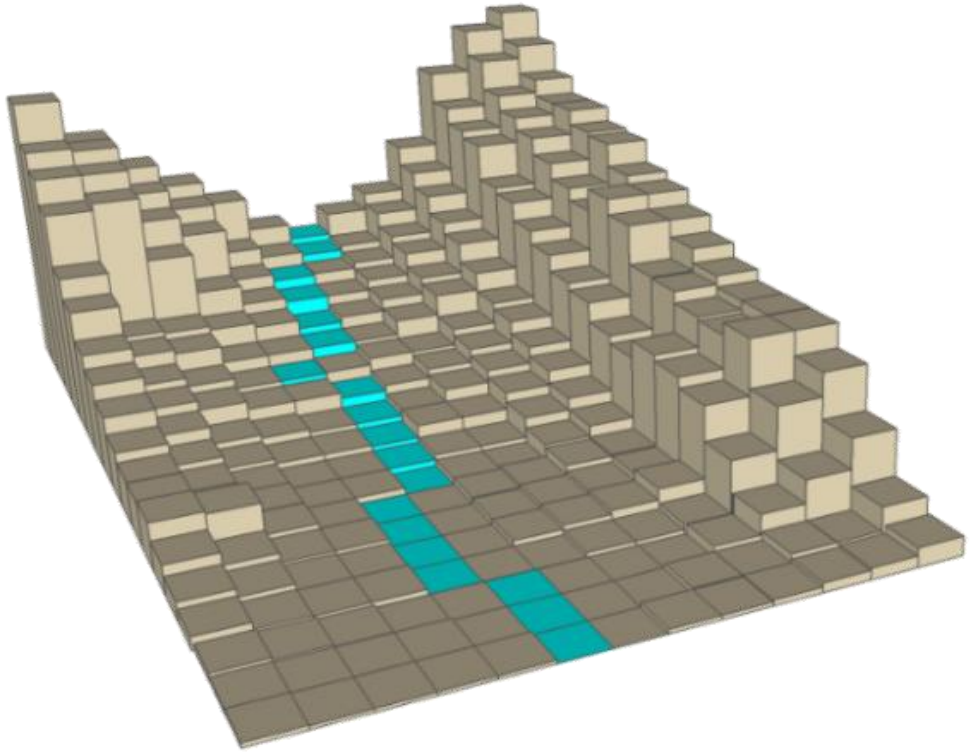
What do you need to run LAHARZ?

- A Digital Elevation Model (DEM) that has an adequate extent (includes the volcano and drainages) and that accurately represents the hydrology



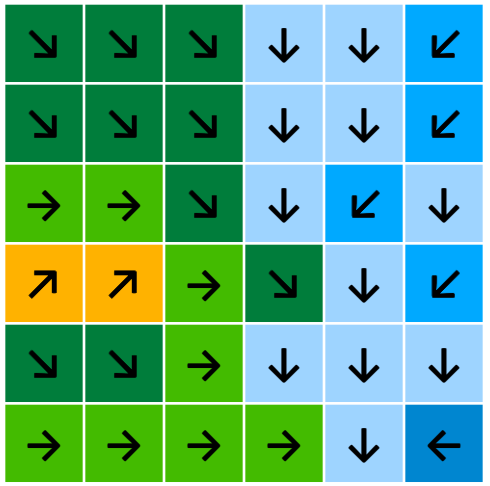
What do you need to run LAHARZ?

- A stream network
- “Create surface hydrology rasters” and “Generate Stream Network” tools in ArcGIS LAHARZ, the “r.stream.extract” in QGIS, or the newest version of the standalone Python LAHARZ
- Generates stream network using the DEM, and a user defined stream threshold
- This is the number of upstream cells that have to flow into a cell to consider it a stream cell. A smaller number here will make a stream network with more branches and longer streams.
- The network should generally match published maps, but should represent the *current* stream network



78	72	69	71	58	49
74	67	56	49	46	50
69	53	44	37	38	48
64	58	55	22	31	24
68	61	47	21	16	19
74	53	34	12	11	12

Elevation (DEM)



Flow direction

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Flow accumulation

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

Stream threshold = 2

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

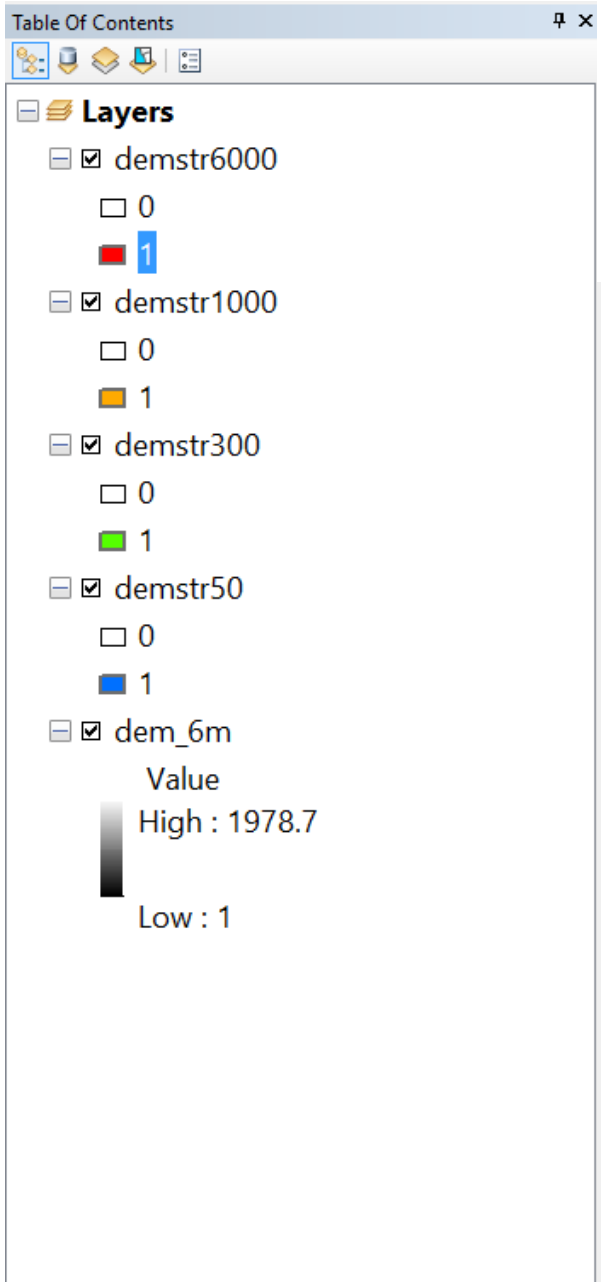
Stream threshold = 6

Lower stream threshold: longer stream network with more branches

Higher stream threshold: shorter stream network with fewer branches

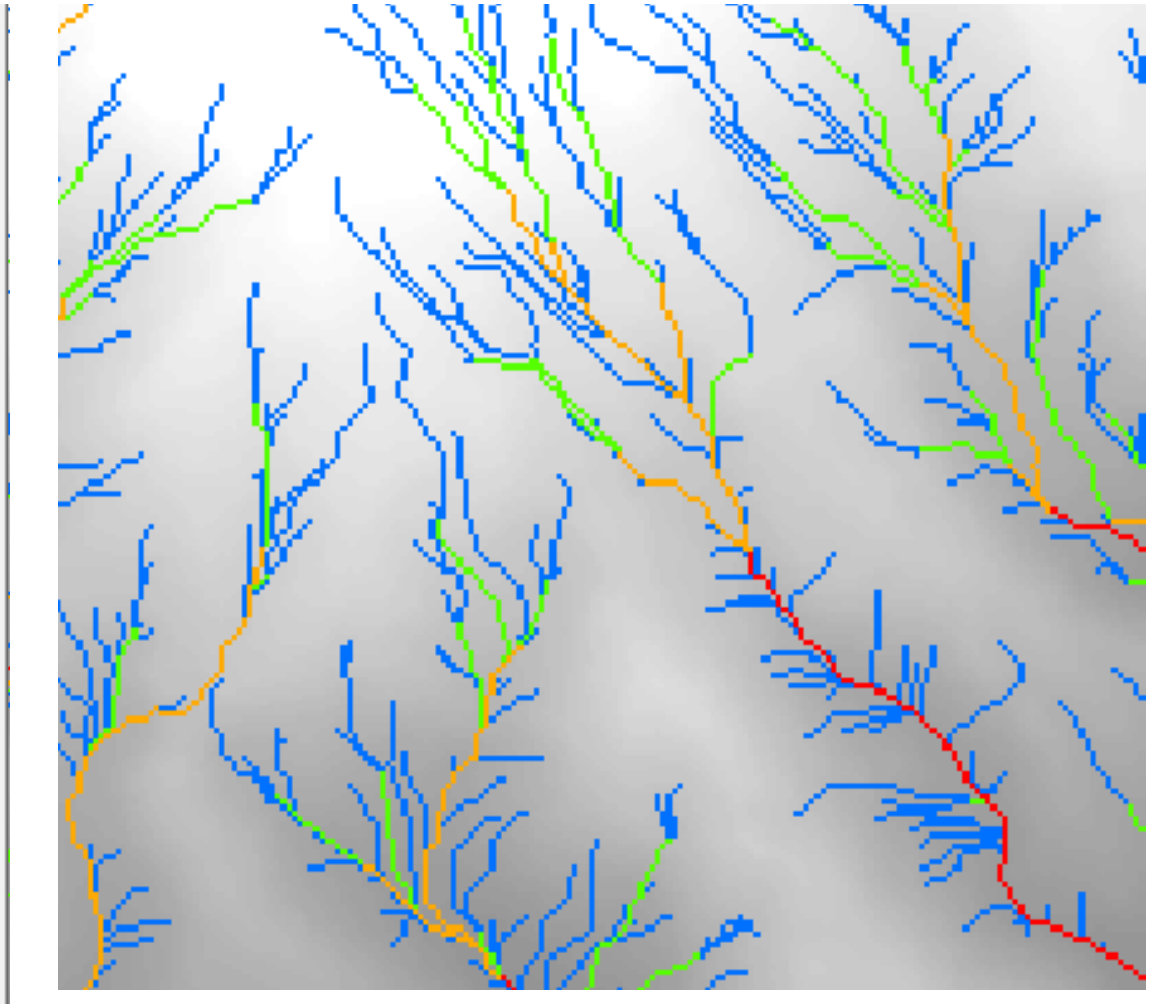
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High stream threshold: shorter stream network with fewer branches

Low stream threshold: longer stream network with more branches

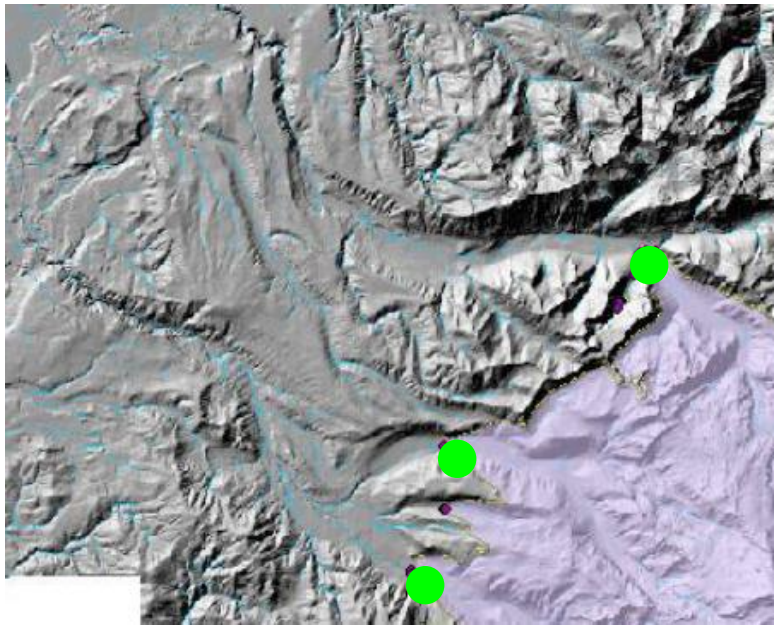


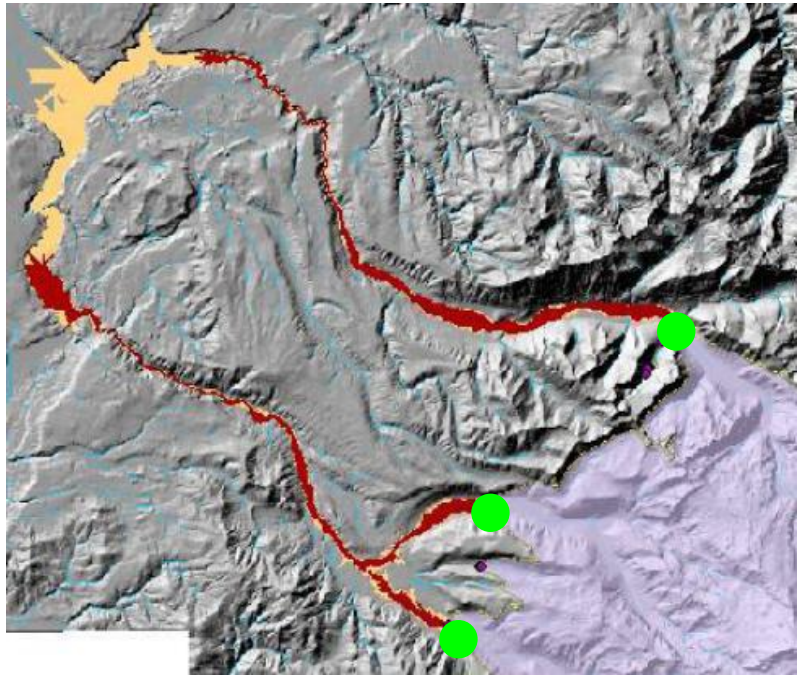
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What do you need to run LAHARZ?

- A starting location or locations:
  - Use the proximal hazard zone tool within LAHARZ to produce an H/L cone that defines the starting locations (can edit these to have fewer points)
  - Manually select starting locations

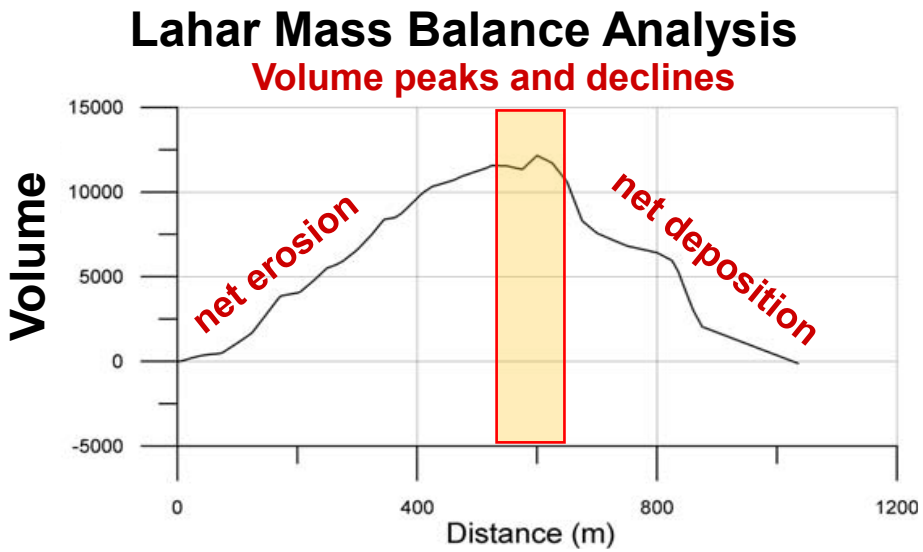




Proximal Hazard Zone for LAHARZ

Why use H/L Cone for lahars?

- Lahars may initiate anywhere on the volcano edifice. We can't know an exact start location in advance.
- Loose pyroclastic flow deposits and ashfall are often the source material for lahars and are most abundant within the H/L cone.
- Lahars gain volume (entrain material) on steep slopes.



- So, we start modeling lahars by estimating where on the flanks the flow volume is greatest.
- We do this by defining an H/L cone that marks the transition between erosion and deposition. That is, inside the H/L cone, the volume is still increasing. Outside the H/L cone, the volume is maintained or decreasing (depositing).
- How do we identify an H/L cone that represents this transition?
    - from net erosion to net deposition
    - from steep slopes to moderate slopes (20°-30°), that is, find the break in slope
    - from pyroclastic debris fans to more channelized flows

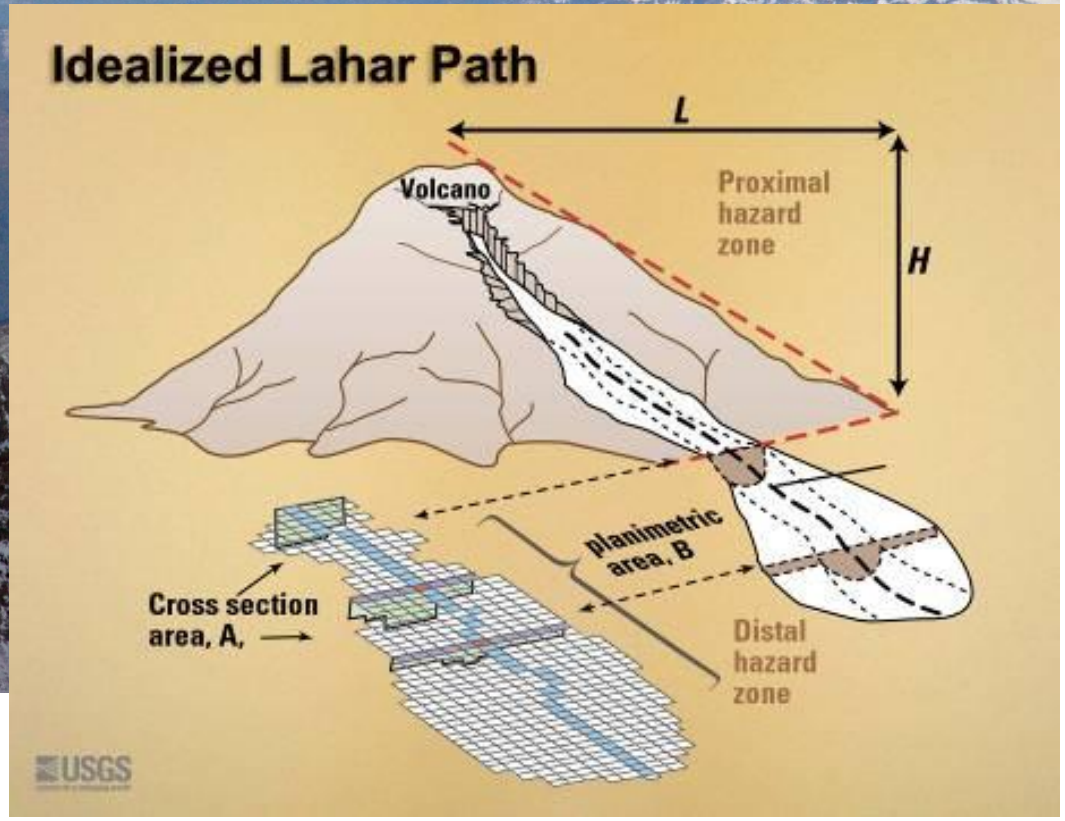
Mt. Rainier, west side

L

V

H

Begin flow hazard-zone computation where deposition first exceeds erosion



Special Cases: not a nice pointy volcano?



Special Cases: not a nice pointy volcano?
Add elevation in LAHARZ



What do you need to run LAHARZ?

- A reasonable range of lahar volumes, considering volcano size, type, history, geology and the goal of the forecast (worst-case scenario vs. most-probable scenario)
- Because of uncertainty and naturally occurring error in the regression equations, you should choose volumes that increase by at least 2x (1, 2, 4, 8, 16 million m^3, for example)
- Even better is to use values that increase by $10^{0.5}$ (1, 3, 10, 30, 100 million m^3, for example)

Summary of key points

1. Documented inundation limits and flow volumes of past lahars provide the data for statistical forecasting of inundation limits for future lahars.

2. Lahar flow volume is the independent variable that must be specified by the user. Generally it's best to posit a range of potential volumes and examine a range of possible outcomes.

3. A rationale must be adopted for establishing starting points for computing inundation limits. For volcanoes, we typically use the intersection of *H/L* cones with valley thalwegs to establish starting points.

Hands-on session to try it out!

Reference slides

Model	Citation	Description/notes	Availability	Count	Percent
LAHARZ (USGS)	Iverson et al. (1998); Schilling (1998); Schilling (2014)	Empirical/statistical model relating volume to planimetric/cross-sectional area	Free, open-source ArcGIS, standalone python, VICTOR	149	71.2
FLO-2D	O'Brien (1993)	2D depth averaged geophysical model; Bingham rheology; designed by engineers for floods, floods in urban environments	Commercial product license $	23	11.0
LaharFlow/MultiHaz	Woodhouse (in preparation); Tierz et al. (2017)	Depth-averaged geophysical model; two-phase, models erosion and deposition	In preparation	6	2.9
HIMAT-FAO	HIMAT-FAO (1987)	?	?	4	1.9
MatHaz	Bertin et al. (2019)	Multi-hazard model; steepest descent/volume-thickness model developed for lava flows, modified for lahars	Free, open-source, available on GitHub	3	1.4
DAMBRK/SMPDBK	Fread (1984); Wetmore & Fread (1991)	Designed for dam-break floods from failed reservoir; models flood peak elevation, travel time using flow routing algorithm	Free, open-source, unclear if still supported	2	0.9
HYDRAUX (USGS)	Hansen (1987); DeLong & Schoellhamer (1989)	1D shallow water geophysical flow model	FORTRAN, open-source, no longer available/supported	2	0.9
VolcFlow	Kelfoun & Druitt (2005)	Geophysical mass flow model developed for granular flows but with multiple rheology options for lahars, one-fluid and two-fluid options	Free, open-source but requires Matlab	2	0.9
Unnamed cost-distance model	Barrantes Castillo & Malavassi Rojas (2015)	GIS-based cost-distance model that determines potential flow paths	Not available, but could implement method in any GIS	1	0.48
Unnamed lahar model	Miyamoto & Ito (2003)	?	?	1	0.48
Unnamed lahar travel time model	Pierson (1998)	Empirical lahar travel time model, does not model inundation	Equations relating flow volume, distance, and time available	1	00.48
Unnamed mass flow model	Mastrolorenzo & Pappalardo (2010)	Geophysical mass flow model developed for granular flows but with multiple rheology options for lahars	Not available	1	0.48

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